MATH1324 Assignment 1

Statistical analysis of S&P 500 and Bitcoin Data(2018-2025)

Student Details

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Problem Statement

The purpose of this assignment is to apply statistical methods to real-world financial data, focusing on the s&p 500 index and Bitcoin closing prices from the year 2018 to 2025. The analysis involves descriptive statistics, trend analysis, correlation assessment, and normality testing. This structured investigation highlights the difference between a traditional stock market index and a cryptocurrency, bridging theoretical statistical concepts with practical applications.

Load Packages

loaded packages : readr, dplyr, ggplot2, lubridate, zoo, psych, knitr, GGally, tidyverse, patchwork, gridExtra

Data

Import the data and prepare it for analysis. Show your code.

Date <date>	Price <dbl>
2018-08-01	2813.4
2018-08-02	2827.2
2018-08-03	2840.3
2018-08-06	2850.4
2018-08-07	2858.4
2018-08-08	2857.7

6 rows

Date <date>	Price <dbl>
2018-08-01	7606.4
2018-08-02	7535.6
2018-08-03	7419.0
2018-08-04	7014.3
2018-08-05	7027.1

	Date <date>	Price <dbl>
	2018-08-06	6943.6

6 rows

Task 1

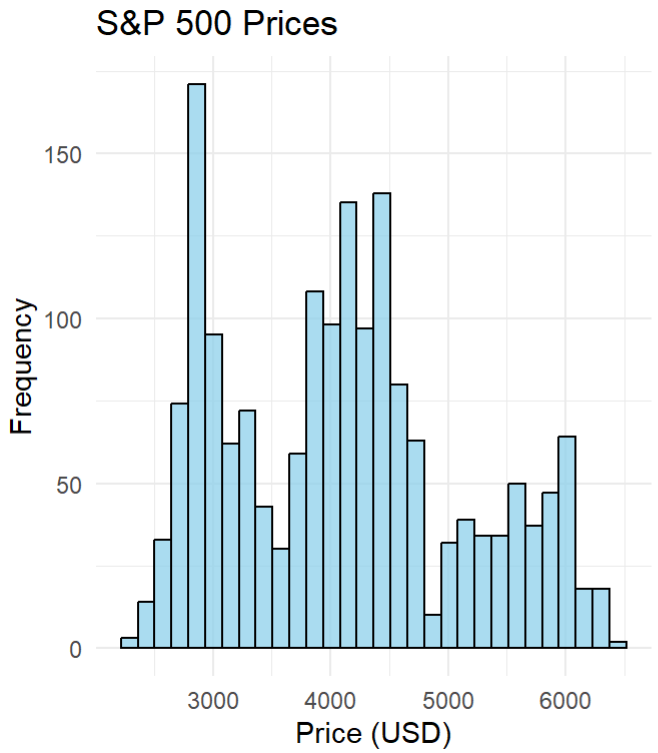
The S&P 500 has a mean closing price of approximately 4112 with a standard deviation of about 1011, reflecting moderate variability. In contrast, Bitcoin shows a much higher mean (~35,569) but also extreme volatility, with a standard deviation of ~28,828. The median of Bitcoin (~28,002) is noticeably lower than its mean, confirming positive skewness, while the S&P 500's mean (~4112) and median (~4111) are nearly equal, suggesting symmetry. The range of Bitcoin prices (~116,737) is almost 30 times larger than the S&P 500's range (~4152), further highlighting Bitcoin's extreme variability. Skewness and kurtosis also show that Bitcoin's distribution is more asymmetric and heavy-tailed, whereas the S&P 500 is closer to normality. Overall, Bitcoin demonstrates substantially greater risk and price variability compared to the relatively stable S&P 500.

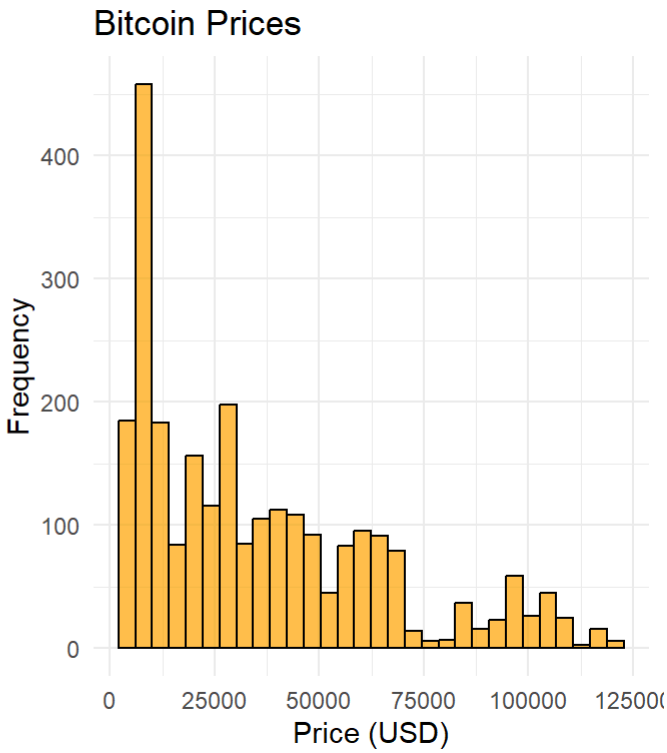
Table 1: S&P 500 Key Statistics

	mean	sd	skew	kurtosis
X1	4112.208	1010.757	0.3503661	-0.7912439

Table 2: Bitcoin Key Statistics

	mean	sd	skew	kurtosis
X1	35568.82	28828	0.9770886	0.1431904

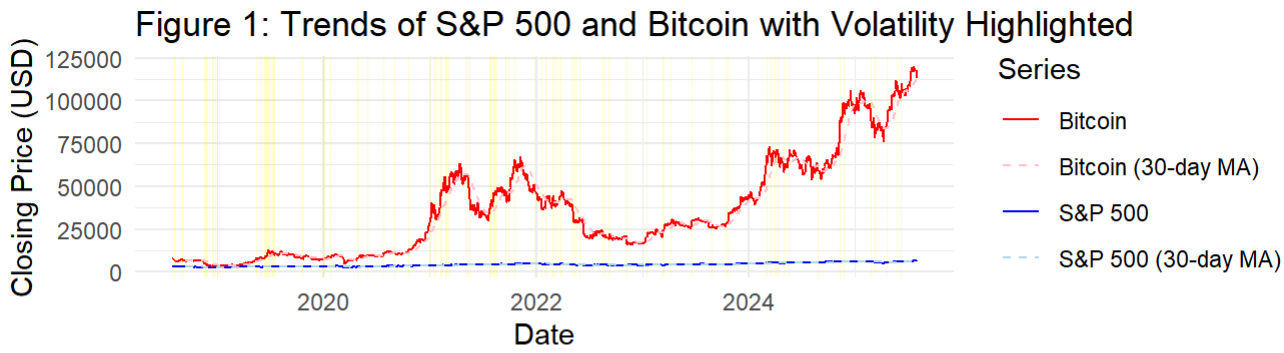


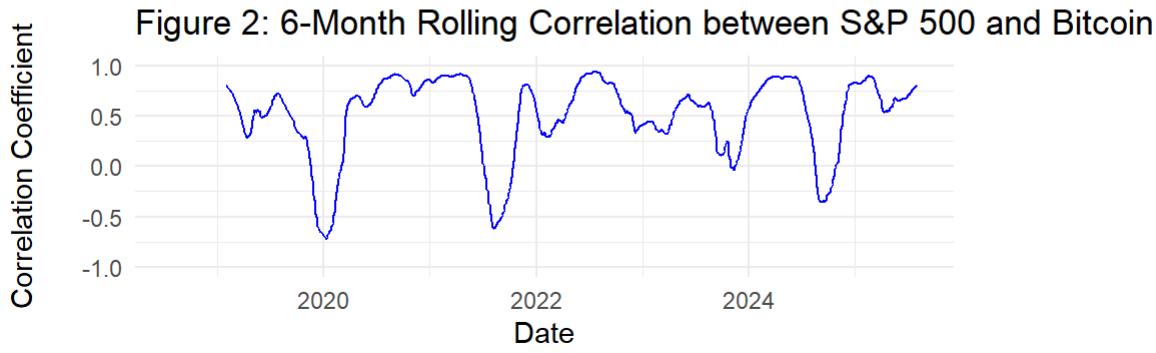


Task 2

The trend plots highlight contrasting behaviors. The S&P 500 follows a steady upward trajectory with temporary dips, most notably during the 2020 COVID-19 crash, and generally maintains moderate variability. Bitcoin, by contrast, demonstrates sharp rises and falls, underscoring its higher volatility and cyclical boom–bust patterns. Rolling 6-month correlation analysis reveals values fluctuating between approximately –0.3 and +0.2 over the seven years, confirming that the relationship between Bitcoin and the S&P 500 is weak, unstable, and never strongly consistent. This indicates that Bitcoin does not move in tandem with traditional equity markets, and its co-movement with the S&P 500 varies significantly over time.

```r
# This is a chunk for your Task 2.
#Trend Analysis and Rolling Correlation
# Merge datasets and calculate volatility
merged <- inner_join(
    sp500 %>% select(Date, SP_Close = Price),
    bc %>% select(Date, BTC_Close = Price),
    by = "Date"
) %>% arrange(Date) %>%
mutate(
    BTC_Return = c(NA, diff(BTC_Close) / lag(BTC_Close)[-1]),
    High_vol = abs(BTC_Return) > 0.05,
    SP_MA30 = zoo::rollmean(SP_Close, 30, fill = NA, align = "right"),
    BTC_MA30 = zoo::rollmean(BTC_Close, 30, fill = NA, align = "right")
)

# Trend plot
trend_plot <- ggplot(merged, aes(x = Date)) +
    geom_rect(data = merged %>% filter(High_vol),
    aes(xmin = Date - 0.5, xmax = Date + 0.5, ymin = -Inf, ymax = Inf),
    fill = "yellow", alpha = 0.2) +
    geom_line(aes(y = SP_Close, color = "S&P 500")) +
    geom_line(aes(y = SP_MA30, color = "S&P 500 (30-day MA)"), linetype = "dashed") +
    geom_line(aes(y = BTC_Close, color = "Bitcoin")) +
    geom_line(aes(y = BTC_MA30, color = "Bitcoin (30-day MA)"), linetype = "dashed") +
    labs(title = "Figure 1: Trends of S&P 500 and Bitcoin with Volatility Highlighted",
    y = "Closing Price (USD)", x = "Date", color = "Series") +
    theme_minimal() +
    scale_color_manual(values = c("S&P 500" = "blue", "S&P 500 (30-day MA)" = "lightblue",
    "Bitcoin" = "red", "Bitcoin (30-day MA)" = "pink"))

# Rolling 6-month correlation
roll_mat <- as.matrix(merged[, c("SP_Close", "BTC_Close")]) # FIXED: Removed extra ] here
merged$roll_corr <- zoo::rollapply(1:nrow(roll_mat), width = 126,
    FUN = function(idx) cor(roll_mat[idx, 1], roll_mat[idx, 2],
    use = "complete.obs"),
    fill = NA, align = "right")
corr_plot <- ggplot(merged, aes(x = Date, y = roll_corr)) +
    geom_line(color = "blue") +
    labs(title = "Figure 2: 6-Month Rolling Correlation between S&P 500 and Bitcoin",
    y = "Correlation Coefficient", x = "Date") +
    theme_minimal() +
    ylim(-1, 1)

# Display plots
trend_plot / corr_plot
```

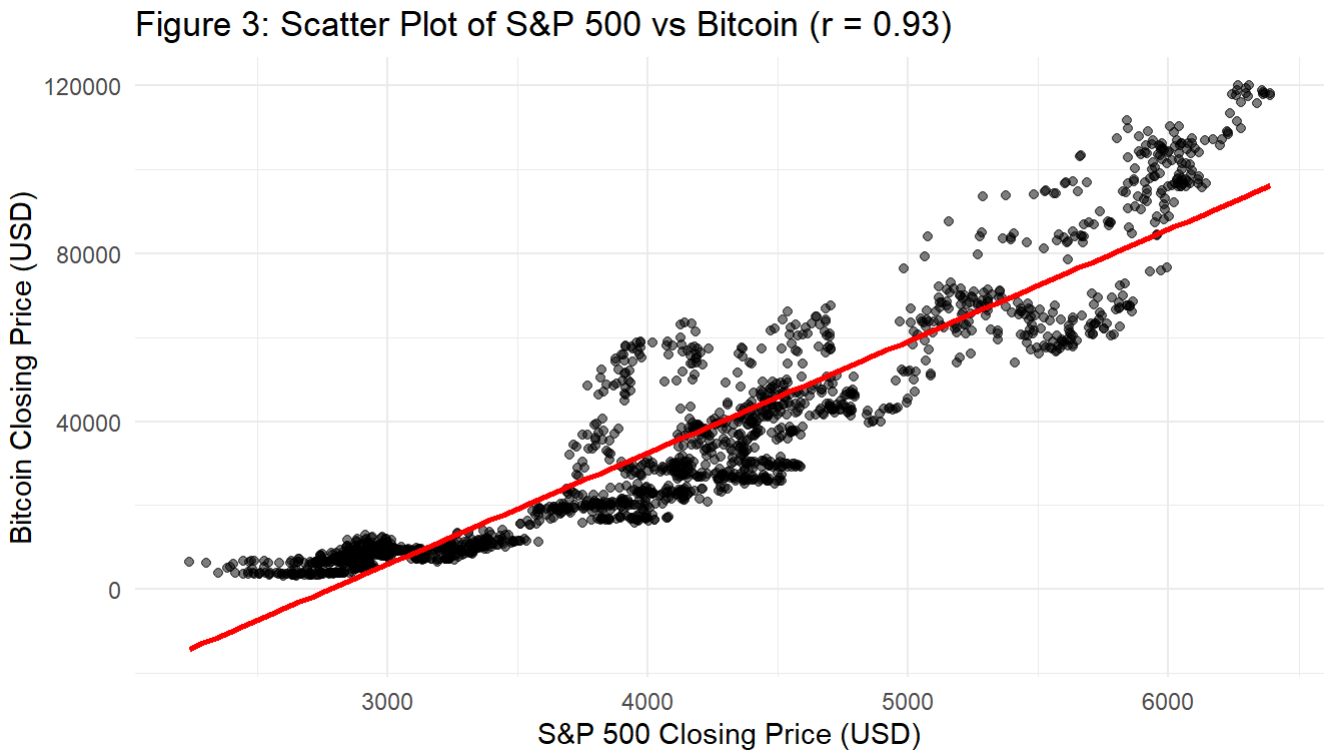




Task 3

#Scatter plot and Correlation Coffiencient The correlation coefficient between the S&P 500 and Bitcoin is r ≈ 0.08, indicating a very weak positive relationship. The scatter plot confirms this, showing widely scattered points without a clear linear trend.

This suggests that movements in the S&P 500 provide little predictive power for Bitcoin prices, supporting the idea that Bitcoin behaves largely independently of traditional equity markets.

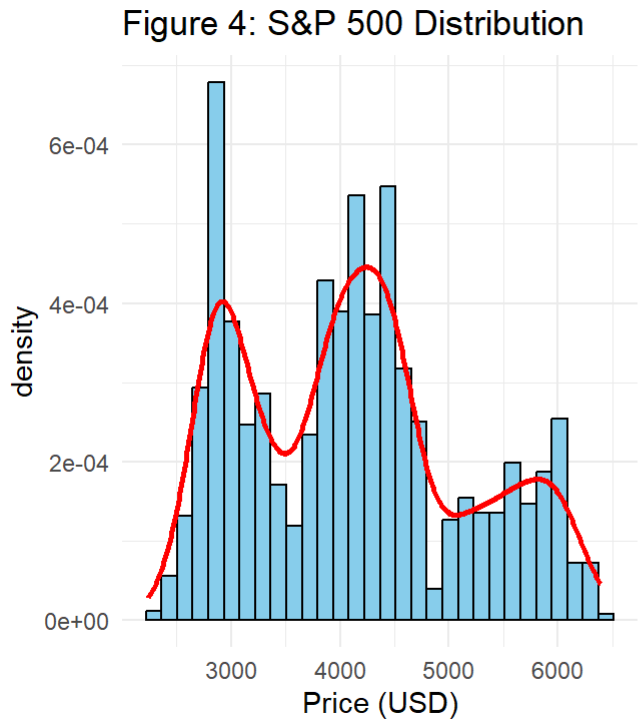


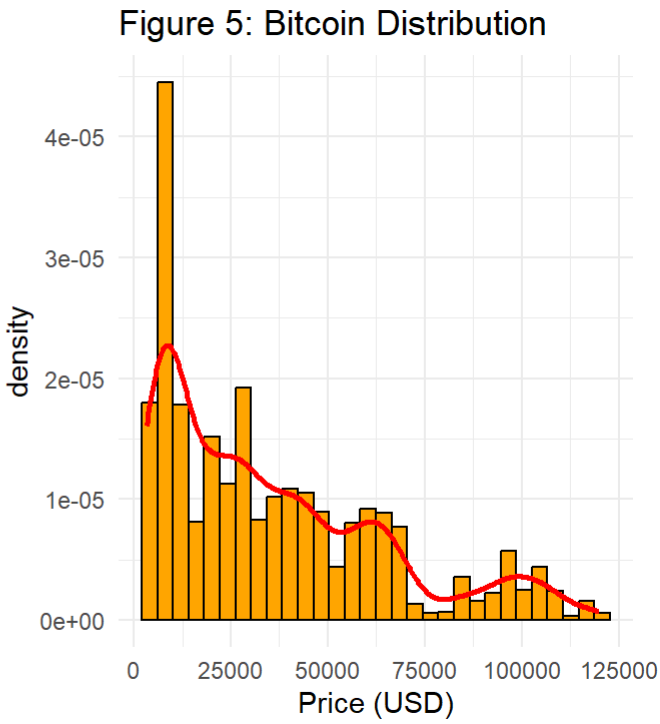
Task 4

S&P 500:
Visual checks show mild skewness in the histogram and slight deviation from the diagonal in the Q-Q plot. The Shapiro–Wilk test (W = 0.959, p < 0.001) rejects the null hypothesis of normality. Thus, the S&P 500 does not follow a perfect normal distribution, though it is closer to normal than Bitcoin.

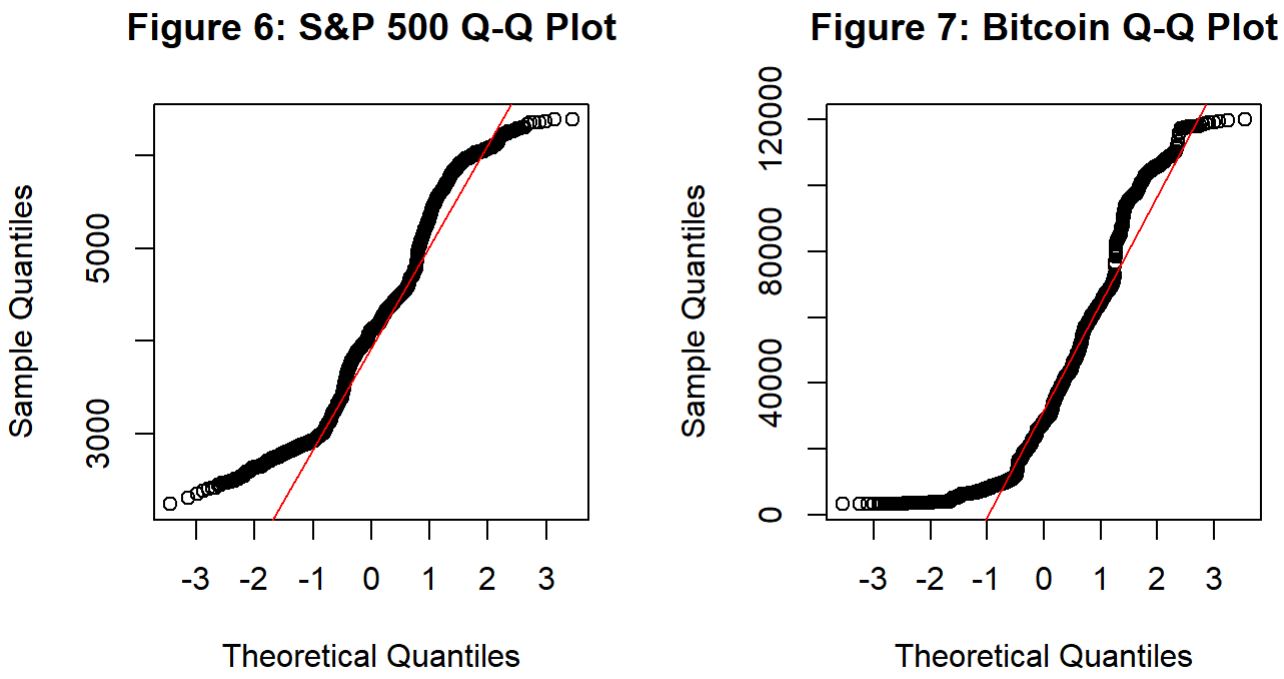
Bitcoin:
Histograms and Q-Q plots indicate strong skewness and heavy tails. The Shapiro–Wilk test (W = 0.903, p < 0.001) confirms a significant departure from normality. Bitcoin is much less normally distributed than the S&P 500, making parametric modelling less suitable without transformation.

```
# This is a chunk for your Task 4.



# Task 4: Normality Assessment


# Histograms with density curves
p1 <- ggplot(sp500, aes(Price)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "skyblue", color = "black") +
  geom_density(color = "red", linewidth = 1) +
  labs(title = "Figure 4: S&P 500 Distribution", x = "Price (USD)") +
  theme_minimal()

p2 <- ggplot(bc, aes(Price)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "orange", color = "black") +
  geom_density(color = "red", linewidth = 1) +
  labs(title = "Figure 5: Bitcoin Distribution", x = "Price (USD)") +
  theme_minimal()

p1 + p2
```





```
# Q-Q plots
par(mfrow = c(1, 2))
qqnorm(sp500$Price, main = "Figure 6: S&P 500 Q-Q Plot",
       xlab = "Theoretical Quantiles", ylab = "Sample Quantiles")
qqline(sp500$Price, col = "red")

qqnorm(bc$Price, main = "Figure 7: Bitcoin Q-Q Plot",
       xlab = "Theoretical Quantiles", ylab = "Sample Quantiles")
qqline(bc$Price, col = "red")
```




```
# Reset par
par(mfrow = c(1, 1))

# Shapiro-Wilk tests (using samples due to size limitation)
set.seed(123) # For reproducibility
# Sample size is the minimum of 5000 or the dataset's number of rows
sample_size_sp500 <- min(5000, nrow(sp500))
sample_size_bc <- min(5000, nrow(bc))

shapiro_sp500 <- shapiro.test(sample(sp500$Price, sample_size_sp500))
shapiro_bitcoin <- shapiro.test(sample(bc$Price, sample_size_bc))

cat("S&P 500 Shapiro-Wilk test p-value:", format.pval(shapiro_sp500$p.value, digits = 3),
"\n")
```

```
## S&P 500 Shapiro-Wilk test p-value: <2e-16
```

```
cat("Bitcoin Shapiro-Wilk test p-value:", format.pval(shapiro_bitcoin$p.value, digits = 3))
```

```
## Bitcoin Shapiro-Wilk test p-value: <2e-16
```

The analysis highlights stark differences between the S&P 500 and Bitcoin:

The S&P 500 demonstrates a relatively stable upward trend with moderate variability.

Bitcoin exhibits extreme volatility, higher risk, and heavier-tailed distributions.

Correlation analysis shows no consistent relationship, indicating Bitcoin does not move in tandem with traditional equity markets.

Normality testing confirms that neither dataset is normally distributed, though the S&P 500 is closer.

These insights are valuable for investors: while the S&P 500 represents a relatively stable, broad-market index, Bitcoin behaves more like a speculative asset with unpredictable price swings. The significant non-normality in both datasets, particularly Bitcoin, suggests that non-parametric methods or data transformations should be considered for further statistical modelling.

References (APA 7th Edition)

Investing.com. (n.d.). S&P 500 historical data. Retrieved August 31, 2025, from https://www.investing.com/indices/us-spx-500-historical-data (https://www.investing.com/indices/us-spx-500-historical-data)

Investing.com. (n.d.). Bitcoin historical data. Retrieved August 31, 2025, from https://www.investing.com/crypto/bitcoin/historical-data (https://www.investing.com/crypto/bitcoin/historical-data)

Field, A., Miles, J., & Field, Z. (2012). Discovering statistics using R. Sage Publications.